

Pruthviraj Mane

Robotics Engineer
System Modeling and Simulation

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GitHub

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
MS(R) (Emobility)	Indian Institute of Technology, Guwahati	7.76	2022
B.Tech. (Mechanical Engg.)	Walchand College of Engineering, Sangli	6.85	2019
Higher Secondary	Maharashtra State Board	84.15%	2015
Secondary	Maharashtra State Board	97.45%	2013

EXPERIENCE

- Project Associate** Aug 2022 - current
AIRL (Artificial Intelligence and Robotics Lab) IISC Bangalore.
 - Worked on modeling and simulation of underactuated autonomous underwater vehicle using MATLAB and Simulink.
 - Worked on developing a ROS-Gazebo based simulator for validation and visualization of 6 DOF vehicle model. This involves path planning, obstacle avoidance, object detection, etc. [GitHub](#)
 - Integrating sensors like IMU, camera, DVL, sonar for collecting synthetic data used for training models.
 - Integrated AUV in Unreal Engine 5 to collect synthetic visual data for underwater vision applications.
 - Developed a 2.5D Forward-Looking sonar-based reactive obstacle avoidance system for AUVs that balances safety and efficiency in 3D navigation.
 - Developed a dynamic ship simulator using Unreal Engine 5 for maritime applications and synthetic data collection.
 - Worked on physics informed neural network with CBF for efficient obstacle avoidance of AUV.

PROJECTS

- Dynamics and stability of electric vehicle (Torque Vectoring)** Jan 2021 - Jul 2022
Dr. B. S. Reddy, Assistant Professor, Dept of Mechanical Engineering IITG
 - Analysed different techniques to obtain differential torque for lateral stability of conventional and electric vehicles.
 - Modelled and compared different control methods like LQR, MPC, SMC to provide differential torque for stability
- Adaptive cruise control for electric vehicle** Jan 2023 - Ongoing
 - Modelled adaptive cruise control system as ADAS in Matlab Simulink for an electric vehicle. [GitHub](#)
 - Used PI controller to maintain the desired speed once the lead vehicle is beyond safe distance.
- Autonomous small scale agricultural plant caretaker** Jun 2018 - May 2019
Prof. R. M. Chanmanwar Assistant Professor, Dept. of Mechanical Engineering WCE Sangli
 - Developed the prototype to automate the farming processes like sowing, moisture sensing and watering.
 - Responsible for modeling the rails and Universal Tool Mount with three tools, on-board seed holder in Solidworks.

TECHNICAL SKILLS

- Programming and Frameworks:** Python, C/C++, ROS, ROS2, Pytorch, MATLAB, Simulink, docker
- Design and Simulation Tools:** Solidworks (CAD), Gazebo, Unreal Engine, Isaac Sim
- Key courses:** Modeling, Dynamics and Control Of EVs, Numerical Methods, Control Systems, Machine Learning

POSITIONS OF RESPONSIBILITY

- Team Captain | Robocon, a national level robotics Competition** Jul 2017 - Mar 2018
 - Led an Interdisciplinary college team of 30 people participating in Robocon 2018 working on a given problem.
- Event Head Futsal | WEST'19 Walchand College Of Engineering Sangli** Jan 2019 - Mar 2019
 - Worked as Event Head Organizer for Futsal Event at WEST'19, a national level sports tournament.

ACHIEVEMENTS

- AIR 38 among 107 teams in Robocon**, As a leader managed the team of 30 students for Robocon 2018 2018
- Scored 98.79 percentile in GATE examination**, GATE 2020 2020

CO-CURRICULAR AND EXTRACURRICULAR ACTIVITIES

- BOSCH's Electric Vehicle Simulation | Inter IIT Tech Meet** 2021
 - Estimated the electric two wheeler parameters for Indian scenario.
- Team ranked among top 60 of the total 1913 Teams | E-Yantra Hackathon 2020: Fighting COVID-19** 2021
 - Worked on online web application for booking of hospital beds during COVID-19
- Produscope Case Study Competition | YouTube | E-Cell IIT Guwahati** 2021
 - As a product manager proposed strategy for unskippable ads with WireFrame and success metrics of the same.